Problem Statement

Manual identification of deceptive URLs has become increasingly difficult due to the rapid growth of internet usage and the increasing number of cyberattacks. Phishing websites often use URLs that closely resemble legitimate websites, making it difficult for users to distinguish between safe and malicious links. Attackers create deceptive URLs by using slight spelling changes, additional characters, misleading domain names, unusual subdomains, and similar-looking words to trick users into believing that a website is genuine. These URLs are commonly shared through emails, text messages, social media platforms, and other online communication channels. When users click on such links, they may be redirected to fraudulent websites designed to steal sensitive information, including usernames, passwords, banking details, and personal data. Even users with basic cybersecurity awareness may fail to identify these deceptive URLs because they can appear trustworthy at first glance.

Traditional methods of identifying malicious URLs often depend on manual inspection, user awareness, and previously reported lists of suspicious websites. However, these approaches may not be sufficient to detect newly created phishing links or URLs that have been modified to avoid detection. Manual verification also requires time and attention, making it unsuitable for analyzing large numbers of URLs quickly and consistently. Therefore, there is a need for an effective and reliable system that can automatically analyze URLs and identify potentially deceptive links based on their characteristics and patterns. Such a system can help distinguish legitimate URLs from suspicious ones, reduce the risk of phishing attacks, and improve online security. By using automated techniques and machine learning algorithms, the proposed system aims to make URL identification faster, more accurate, and more efficient, helping users make safer decisions while browsing the internet.